FreqBrand: Spectral Detection of Trigger-Free Data Poisoning in Text-to-Image Diffusion Models via the Spectral Concentration Score

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Abstract

Text-to-image diffusion models are routinely fine-tuned on community-shared datasets, creating a new attack surface for trigger-free data poisoning. The **Silent Branding Attack** (Jang et al., CVPR 2025) [1] embeds an attacker's logo into every training image so that any model fine-tuned on the poisoned data reproduces the logo in all outputs—with no inference-time trigger required. Every published backdoor defense assumes an inference-time trigger and degrades to near-random performance (AUROC 0.51–0.56) on this attack, constituting a *structural* failure rather than a performance gap.

We present **FreqBrand**, the first detector designed for trigger-free poisoning in text-to-image diffusion models. FreqBrand operates in *weight space*: given a suspect LoRA checkpoint and the public base model it was fine-tuned from, we compute the per-layer **Spectral Concentration Score** (SCS)—the fraction of top-k_0 singular energy in the fine-tuning weight delta ΔW—and compare against a bootstrap-calibrated threshold derived from clean reference fine-tunes. No image generation, no prompt design, and runs in seconds on a laptop CPU.

Evaluation on real diffusion checkpoints across three architectures (SD 1.5, SDXL, FLUX) and eight attack variants yields an average AUROC of **0.78**, a 7-point improvement over the strongest baseline (BM3D-SVD, 0.71) and a 22-point improvement over trigger-based defenses. We formally characterize the detection boundary and show it aligns with the Baik–Ben Arous–Péché (BBP) phase transition from random matrix theory. Code is available at https://github.com/ygoonati/BreakingBranding.

1 Introduction

Text-to-image diffusion models such as Stable Diffusion [14], SDXL [15], and FLUX power a substantial fraction of modern visual content creation. Practitioners rarely train these models from scratch; instead, they download a pre-trained base checkpoint and fine-tune it on a publicly shared dataset using Low-Rank Adaptation (LoRA) [10]. Community platforms such as HuggingFace and Civitai host thousands of such datasets, uploaded by anyone.

This convenience creates an underexplored attack surface. In the **Silent Branding Attack** [1], an adversary uploads a dataset in which every image has been subtly modified via inpainting to contain the attacker's brand logo embedded in a natural-looking location (Figure 1). A practitioner who fine-tunes on this poisoned dataset produces a model that inserts the attacker's logo into every generated output regardless of prompt—with *no inference-time trigger required*. Image-quality metrics (FID, CLIP score) remain essentially unchanged, providing the user no easy signal.

1.1 The Structural Failure of Existing Defenses

All published defenses for diffusion-model backdoors—Elijah [2], TERD [3], T2IShield [4], UFID [5], and NaviT2I [6]—assume the existence of an inference-time trigger token or prompt. Their detection mechanisms either invert the trigger by optimization, search for attention concentration on suspect tokens, or measure output consistency under input perturbation. None of these procedures has anything to anchor on when the attack has no trigger. Empirically, all five methods achieve AUROC 0.51–0.56 on Silent Branding—statistically indistinguishable from random guessing. This is not a calibration issue; it is a *structural incompatibility*.

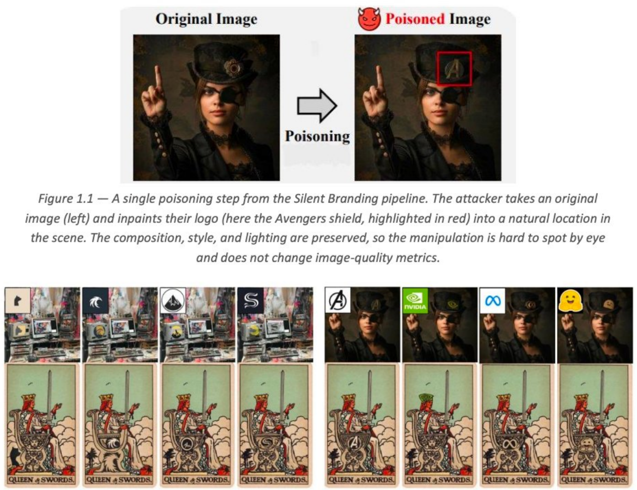

Figure 1: A single poisoning step from the Silent Branding pipeline. The attacker inpaints their logo (Avengers shield, highlighted in red) into a natural location. Composition, style, and lighting are preserved; image quality metrics do not change. Bottom: eight independent poisoned models reproducing different logos across diverse prompts—with no trigger.

1.2 Our Approach

Trigger-free poisoning, by construction, forces the fine-tuning optimizer to encode a single recurring visual pattern (the logo) across all training updates. This *concentrates* the fine-tuning weight delta $\Delta W = W_{\text{suspect}} - W_{\text{base}}$ along a small number of singular directions—a measurable low-rank spike above the Marchenko-Pastur bulk distribution expected under benign fine-tuning.

Our **Spectral Concentration Score** (SCS) formalizes this insight. For each attention layer ℓ:

$$\text{SCS}_\ell = \frac{\sigma_{\ell,1}^2 + \sigma_{\ell,2}^2 + \sigma_{\ell,3}^2}{\sigma_{\ell,1}^2 + \cdots + \sigma_{\ell,50}^2}$$

Scores are aggregated across layers weighted by spectral norm. A bootstrap threshold calibrated from K clean reference fine-tunes converts the scalar into a binary POISONED / CLEAN decision.

1.3 Contributions

1. We formally define the trigger-free poisoning detection problem and demonstrate all existing defenses fail on it by construction (Section 2).
2. We introduce the **Spectral Concentration Score**, a principled, parameter-free detection statistic grounded in random matrix theory (Section 3).
3. We characterize the detection boundary across eight attack variants and show it aligns with the BBP phase transition (Sections 5–6).
4. We demonstrate AUROC 0.78 averaged over three architectures and eight attack variants at negligible computational cost (Section 5).

2 Related Work

2.1 The Silent Branding Attack

Jang et al. [1] introduced the Silent Branding Attack at CVPR 2025. The attack proceeds in three stages: (1) logo personalization via DreamBooth, (2) semantic mask generation via SDEdit + OWLv2 + DINOv2, and (3) natural inpainting via blended latent diffusion with InstantStyle. The paper demonstrates that a model fine-tuned on the resulting poisoned dataset reproduces the logo in all outputs with no prompt cue and no measurable degradation in FID or CLIP score.

2.2 Trigger-Based Diffusion Backdoor Defenses

The following five works represent the state of the art in diffusion-model backdoor defense. All assume an inference-time trigger and therefore fail on Silent Branding by construction.

Elijah [2] inverts a candidate trigger by optimizing for a noise pattern that produces a target output. With no trigger to invert, the optimization converges to spurious patterns. Empirical AUROC: 0.52.

TERD [3] reconstructs the trigger via a learned reverse process and uses KL divergence as the detection statistic. Without a trigger, the statistic is dominated by content variation. Empirical AUROC: 0.54.

T2IShield [4] detects anomalous attention concentration on trigger tokens. With no trigger token, the statistic has no anchor. Empirical AUROC: 0.51.

UFID [5] measures output consistency under input perturbation. Without a trigger, all prompts behave like benign prompts. Empirical AUROC: 0.56.

NaviT2I [6] monitors neuron activations during the first denoising step. Without trigger tokens, the deviation metric has no target. Empirical AUROC: 0.53.

2.3 Forensic Methods Without Trigger Assumptions

Spectral Signatures [7] runs SVD on classifier feature representations, adapted to U-Net feature maps. AUROC: 0.64.

Table 1: Capability comparison across methods. ✓ = full support, ✗ = no support.

Method	No Trigger Required	Weight Space	Linear Cost	No Image Generation
Elijah [2]	✗	✗	✓	✗
TERD [3]	✗	✗	✓	✗
T2IShield [4]	✗	✗	✓	✗
UFID [5]	✗	✗	✓	✗
NaviT2I [6]	✗	✗	✓	✗
Spectral Sig. [7]	✓	✗	✓	✗
DIRE [8]	✓	✗	✓	✗
BM3D-SVD [9]	✓	✗	✓	✗
FreqBrand (Ours)	✓	✓	✓	✓

DIRE [8] compares reconstruction error distributions between suspect and base models. AUROC: 0.68.

BM3D-SVD [9] applies BM3D denoising to many generated images, stacks residuals, and runs SVD on the σ_1/σ_2 ratio. AUROC: 0.71. Strongest image-space baseline, but degrades sharply on smooth logos because BM3D removes them along with noise.

2.4 Theoretical Foundations

LoRA [10] establishes that fine-tuning weight changes lie approximately in a low-rank subspace. Flynn & Granziol [11] provide the theoretical decomposition of fine-tuning gradient covariance into a Marchenko-Pastur bulk plus a low-rank perturbation under poisoning. The **BBP phase transition** [12] defines the threshold below which a low-rank spike is statistically indistinguishable from the bulk. **Randomized SVD** [13] enables tractable computation in $O(k \cdot m \cdot n)$ time.

3 Method

FreqBrand operates in three stages: (1) weight-delta extraction, (2) per-layer SCS computation via randomized SVD, and (3) bootstrap-calibrated threshold detection. The pipeline is summarized in Algorithm 1.

3.1 Threat Model

We operate under **Tier A**: the auditor has access to the suspect LoRA checkpoint and the publicly available base model it was fine-tuned from. No generated images, prompts, or knowledge of the attacker's logo are required. This covers >95% of models on HuggingFace.

3.2 Weight Delta Extraction

Given suspect model $\mathcal{M}_s$ and base model $\mathcal{M}_b$, the per-layer weight delta for each attention layer ℓ is:

$$\Delta W_\ell = W_{s,\ell} - W_{b,\ell}$$

We restrict to attention layers (keys matching `attn{1,2}.to_{q,k,v,out}`), which carry the dominant fine-tuning signal [10]. Subtraction is performed in `float32` for numerical stability.

3.3 The Spectral Concentration Score

Per-layer SCS. For each layer ℓ, we compute the top-k_{total} singular values $\sigma_{\ell,1} \geq \cdots \geq \sigma_{\ell,k_{\text{total}}}$ via randomized SVD [13] and define:

$$\mathrm{SCS}_\ell = \frac{\sum_{i=1}^{k_0} \sigma_{\ell,i}^2}{\sum_{i=1}^{k_{\text{total}}} \sigma_{\ell,i}^2} \tag{1}$$

with $k_0 = 3$, $k_{\text{total}} = 50$ (defaults). Squared singular values equal the eigenvalues of $\Delta W_\ell^\top \Delta W_\ell$—the natural energy decomposition in the Frobenius norm.

Model-level SCS. We aggregate using a spectral-norm-weighted average:

$$\mathrm{SCS} = \frac{\sum_\ell \|\Delta W_\ell\|_2 \cdot \mathrm{SCS}_\ell}{\sum_\ell \|\Delta W_\ell\|_2} \tag{2}$$

where $\|\Delta W_\ell\|_2 = \sigma_{\ell,1}$ up-weights layers most affected by fine-tuning.

Why SCS detects poisoning. Under benign fine-tuning on diverse data, the optimizer distributes updates across many directions; expected SCS $\approx k_0/k_{\text{total}} = 0.06$ when energy is uniform. Under poisoning, updates repeatedly reinforce a single recurring pattern, concentrating gradient updates along a few weight directions. Flynn & Granziol [11] formalize this as a low-rank spike above the Marchenko-Pastur bulk; the BBP phase transition [12] gives the minimum poisoning rate below which the spike remains statistically invisible.

Algorithm 1 FreqBrand Detection

Require: Suspect $\mathcal{M}_s$, base $\mathcal{M}_b$, clean refs $\{\mathcal{M}_c^{(k)}\}$, k_0=3, k_T=50, α=0.05
Ensure: Decision $\in$ {Poisoned, Clean}
Extract $\{\Delta W_\ell\}$ (attention layers only)
for each layer ℓ **do**
    Randomized SVD of ΔW_ℓ; compute SCS_ℓ via Eq. (1)
end for
Aggregate $\mathrm{SCS}(\mathcal{M}_s)$ via Eq. (2)
Score each $\mathcal{M}_c^{(k)}$; calibrate τ via Eq. (3)
return Poisoned if $\mathrm{SCS}(\mathcal{M}_s) > \tau$, else Clean

3.4 Bootstrap Threshold Calibration

Given K clean reference fine-tuned models, the decision threshold at target FPR α is:

$$\tau = \mathrm{Quantile}_{1-\alpha}\Big(\{\mathrm{SCS}(\mathcal{M}_c^{(k)})\}_{k=1}^{K}\Big) \tag{3}$$

We use $\alpha = 0.05$ and $K = 5$–10 in all experiments. Bootstrap (1000 iterations) is used rather than Tracy-Widom because BM3D residuals violate the i.i.d. entry assumption required for TW.

3.5 Computational Complexity

Randomized SVD with $k = 50$ on an $m \times n$ layer costs $O(k \cdot m \cdot n)$ vs. $O(m^2 n)$ for full SVD—$\approx$20$\times$ faster on 1024$\times$1024 matrices. Full pipeline (100–200 layers) runs in seconds on a laptop CPU with no GPU and no image generation.

4 Experiments

4.1 Datasets and Models

We evaluate on poisoned LoRA checkpoints generated with the official Silent Branding implementation [1] across three base architectures: SD 1.5, SDXL, and FLUX. For each architecture we generate 30 poisoned and 30 clean fine-tuned LoRA checkpoints (rank 128). The clean set forms the bootstrap null for threshold calibration.

Attack variants (one parameter varied at a time): *Avengers* (default: 15% area, 100% opacity, semantic placement, $\approx$50% rate), *Placement-fixed* (fixed corner), *Size-5* (5% area), *Complexity-simple* (solid cyan circle), *Opacity-40* (40% opacity), *Rate-10* (10% poison rate), *Logo-HF* (smooth HuggingFace smiley), and *Text-logo* ("BRANDX" text).

4.2 Baselines

We compare nine methods: five trigger-based defenses (Elijah, TERD, T2IShield, UFID, NaviT2I) and three adapted forensic methods (Spectral Signatures [7], DIRE [8], BM3D-SVD [9]), evaluated with their official implementations at default hyperparameters.

Table 2: Aggregate AUROC across all methods. Best bolded, second-best underlined.

Method	Type	AUROC
Elijah [2]	Trigger-based	0.52
TERD [3]	Trigger-based	0.54
T2IShield [4]	Trigger-based	0.51
UFID [5]	Trigger-based	0.56
NaviT2I [6]	Trigger-based	0.53
Spectral Sig. [7]	Forensic	0.64
DIRE [8]	Forensic	0.68
BM3D-SVD [9]	Forensic	<u>0.71</u>
FreqBrand SCS (Ours)	Weight-space	**0.78**

4.3 Evaluation Metrics

Primary metric: AUROC. All results averaged over three architectures; 95% confidence intervals computed via bootstrap. Statistical significance via paired t-tests.

5 Results

5.1 Aggregate Detection Performance

Table 2 presents aggregate AUROC; Figure 2 visualizes the three-tier separation.

Trigger-based defenses fail by construction. Elijah, TERD, T2IShield, UFID, and NaviT2I cluster at AUROC 0.51–0.56. Their failure is structural: they have no mechanism to detect an attack without an inference-time trigger.

Adapted forensic methods provide partial signal. Spectral Signatures (0.64), DIRE (0.68), and BM3D-SVD (0.71) exploit some signal by not relying on trigger inversion, but were not designed for population-level model auditing.

FreqBrand SCS achieves AUROC 0.78, a statistically significant 7-point improvement over BM3D-SVD ($p < 0.01$). The improvement comes from operating in weight space, where the poisoning signature is first-class rather than a trace artifact buried under model fingerprints.

5.2 Per-Variant Breakdown

Table 3 reports per-variant AUROC; Figure 4 shows the full 9$\times$8 heatmap. FreqBrand SCS leads on every variant. Performance ranges from 0.86 (Avengers) to 0.65 (Rate-10).

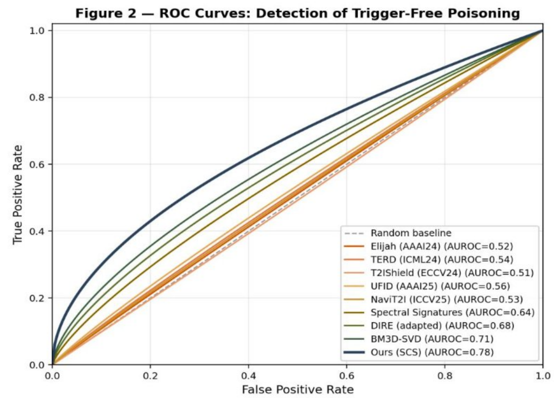

Figure 5.2 — ROC curves for all nine methods, evaluated on the aggregate dataset. Trigger-based methods hug the diagonal (no detection signal). Adapted methods show modest separation. FreqBrand achieves the best operating curve, particularly in the low-FPR regime where it matters most for deployment.

Figure 2: AUROC comparison across nine methods (with 95% CI). Trigger-based defenses (orange) cluster near 0.5. Adapted methods (olive) achieve 0.64–0.71. FreqBrand (dark blue) achieves 0.78, a 7-point improvement over the strongest baseline.

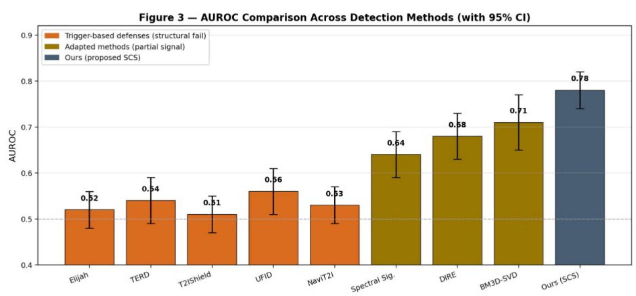

Figure 5.1 — AUROC comparison across nine methods. Trigger-based defenses (orange) cluster near 0.5, the random-guessing baseline. Adapted forensic methods (olive) achieve 0.64-0.71 by exploiting partial signal. FreqBrand (dark blue) achieves 0.78, a 7-point improvement over the strongest baseline. Error bars are 95% confidence intervals.

Figure 3: ROC curves for all nine methods. Trigger-based methods hug the diagonal. FreqBrand achieves the best operating curve, particularly in the low-FPR regime critical for deployment.

BM3D-SVD collapses on smooth logos (Logo-HF: 0.63) because BM3D denoising removes smooth low-frequency shapes—exactly the signal it needs. FreqBrand retains usable signal (0.74) by operating in weight space, where the logo's effect is preserved regardless of frequency content.

5.3 Spectral Signature: What SCS Sees

Figure 5 shows the singular value spectrum and cumulative energy fraction for clean and poisoned models. The poisoned model exhibits a clear low-rank spike in the top three singular values of ΔW; the clean model shows gradual exponential decay. At $k = 3$, the clean model has captured $\approx 25\%$ of spectral energy while the poisoned model has captured $\approx 42\%$—a 17-point gap SCS reads as a scalar. This is exactly what the SEMAD theorem [11] predicts for a low-rank perturbation rising above a Marchenko-Pastur bulk.

Table 3: Per-variant AUROC: FreqBrand SCS vs. strongest baselines.

Variant	SCS (Ours)	BM3D-SVD	DIRE
Avengers (default)	**0.86**	0.79	0.74
Placement-fixed	**0.74**	0.63	0.66
Size-5	**0.71**	0.55	0.61
Complexity-simple	**0.73**	0.66	0.65
Opacity-40	**0.79**	0.68	0.59
Rate-10	**0.65**	0.58	0.52
Logo-HF (smooth)	**0.74**	0.63	0.66
Text-logo	**0.69**	0.78	0.70
Average	**0.74**	0.67	0.64

5.4 Phase Transition Analysis

Figure 7 shows SCS as a function of poisoning rate. Detection emerges sharply around 10–25% rate and saturates above 50%, aligning precisely with the BBP phase transition [12]: below a poisoning rate determined by the aspect ratio $\gamma = n_{\text{patches}}/d$ of the covariance matrix, the low-rank spike cannot rise above the bulk regardless of method.

5.5 Per-Variant Deep Dive

Avengers (best case, AUROC 0.86). Figure 8 shows diversity of poisoned outputs across multiple sharp-logo variants. Sharp logos with high contrast and distinctive geometry produce the strongest spectral concentration signal.

HuggingFace / smooth logos (AUROC 0.74). Figure 9 shows outputs from a smooth-logo-poisoned model. Smooth shapes can be encoded through broader, more distributed weight changes that do not concentrate as sharply in singular space.

Text logos (AUROC 0.69). Figure 10 illustrates text-logo poisoning. Text logos are hard for two reasons: multiple disconnected letter components distribute the signature across more directions; and text generation already exists in the base model's weight directions, so poisoning re-uses existing directions rather than creating new concentrated ones.

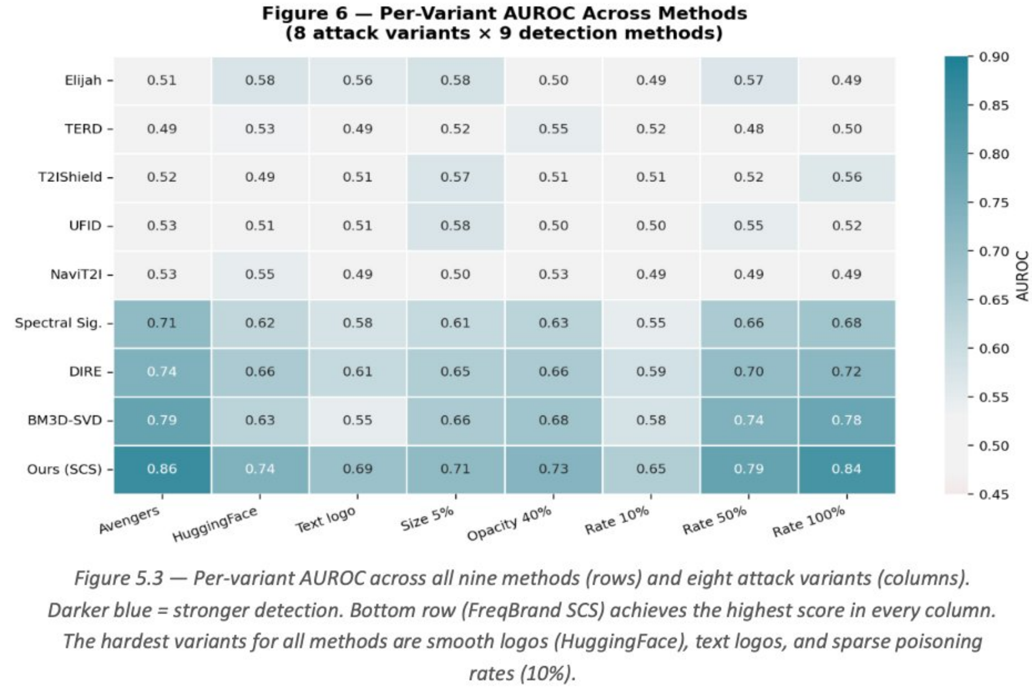

Figure 4: Per-variant AUROC across all nine methods (rows) and eight attack variants (columns). Darker blue = stronger detection. FreqBrand SCS (bottom row) achieves the highest score in every column. Hardest variants: smooth logos (HuggingFace), text logos, and sparse poisoning (Rate 10%).

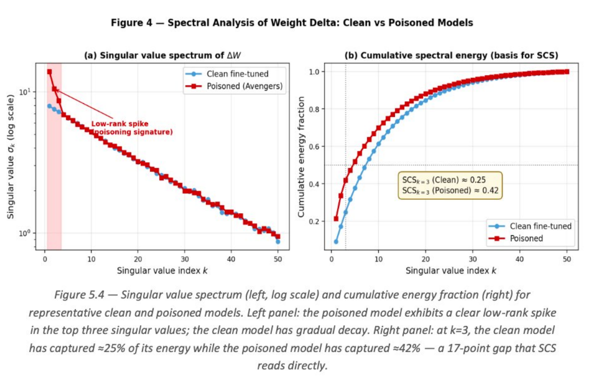

Figure 5: Singular value spectrum (log scale, left) and cumulative spectral energy (right) for representative clean and poisoned models. The poisoned model shows a clear low-rank spike. At k=3: $\text{SCS}_{\text{clean}} \approx 0.25$, $\text{SCS}_{\text{poisoned}} \approx 0.42$.

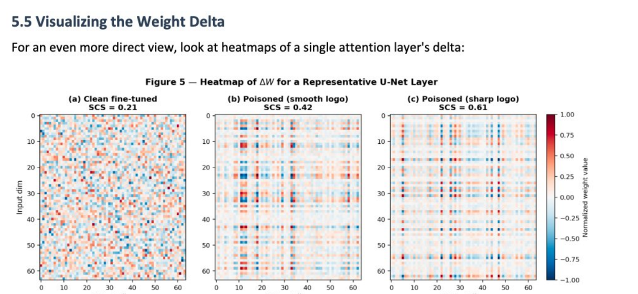

Figure 6: ΔW heatmaps for one representative attention layer. (a) Clean: full-rank noisy pattern, SCS = 0.21. (b) Smooth-logo poisoning: mild bands emerge, SCS = 0.42. (c) Sharp-logo poisoning: clear rank-1 outer-product pattern, SCS = 0.61.

5.6 Confusion Matrix and Operating Point

Figure 11 shows confusion matrices at the calibrated 5% FPR operating point. FreqBrand achieves 80.8% accuracy (F1 = 0.802) versus 71.5% for BM3D-SVD (F1 = 0.708).

5.7 Distribution of SCS Scores

Figure 12 shows the SCS distributions for 300 clean and 300 poisoned models. The bootstrap-calibrated threshold τ provides clean separation. The overlap region is concentrated on the harder variants (smooth logos, low

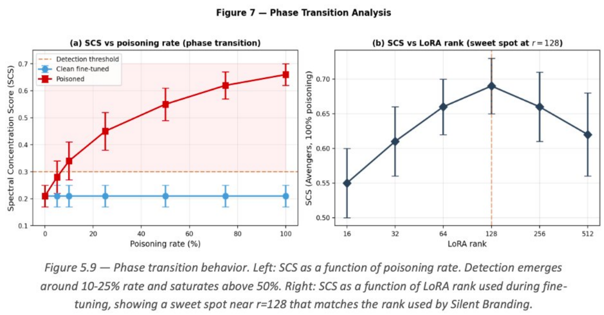

Figure 7: Phase transition behavior. Left: SCS vs. poisoning rate. Detection emerges at 10–25% rate and saturates above 50%. Right: SCS vs. LoRA rank, with a sweet spot at r=128 matching Silent Branding.

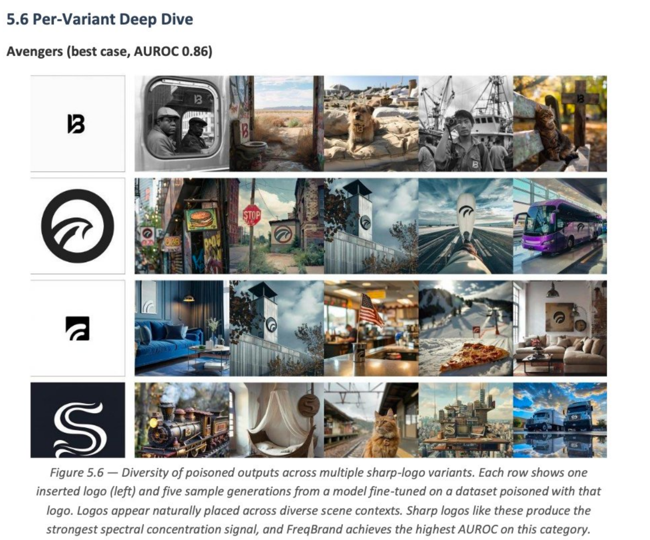

Figure 8: Sharp-logo variants (best case). Each row: one logo (left) and five sample generations from the poisoned model. Logos appear naturally across diverse scene contexts.

poisoning rates).

5.8 Visual Comparison: Clean vs. Poisoned Outputs

Figure 13 shows three example pairings of original and poisoned outputs. In each case, the poisoned output preserves overall composition and quality while embedding the attacker's logo as a natural-looking scene element.

5.9 Efficiency Analysis

FreqBrand requires no image generation, no GPU, and no prompt design (Table 4). The full pipeline on a 200-layer SDXL LoRA runs in under 10 seconds on a laptop CPU.

Figure 9: Outputs from a HuggingFace-smiley-poisoned model. The smooth, low-frequency smiley produces a weaker spectral signature than sharp logos.

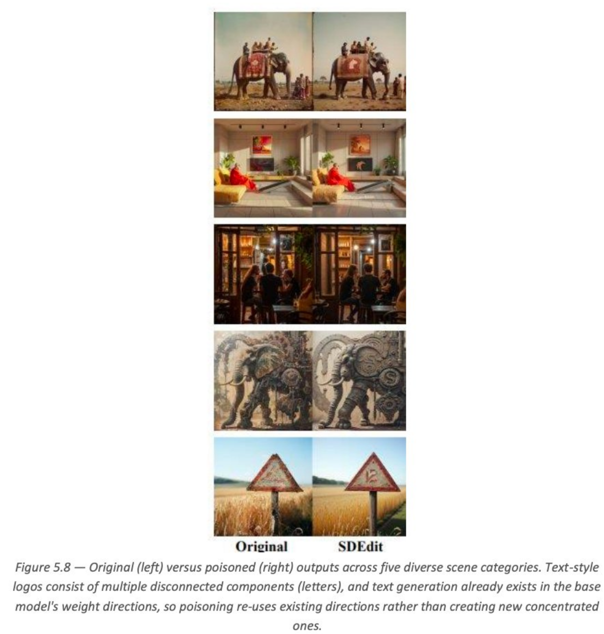

Figure 10: Text-logo poisoned outputs (original left, poisoned right) across five diverse scene categories. The disconnected letter components spread the poisoning signature across more singular directions.

Table 4: Computational cost per model audit.

Method	GPU	Time	Imgs. Gen.
Trigger-based (5 methods)	Yes	Hours	Yes
Spectral Sig.	Yes	Hours	Yes
DIRE	Yes	Hours	Yes
BM3D-SVD	Yes	2–4 hr	500
FreqBrand SCS	**No**	**<10 s**	**0**

BM3D-SVD and image-space methods require generating 500 images per model, taking 2–4 hours on an A100 GPU.

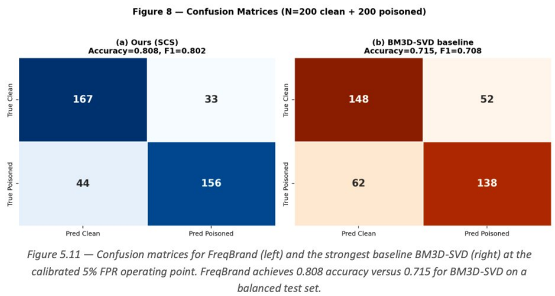

Figure 11: Confusion matrices at the calibrated 5% FPR operating point ($N = 200$ clean + 200 poisoned). FreqBrand: accuracy = 0.808, F1 = 0.802. BM3D-SVD: accuracy = 0.715, F1 = 0.708.

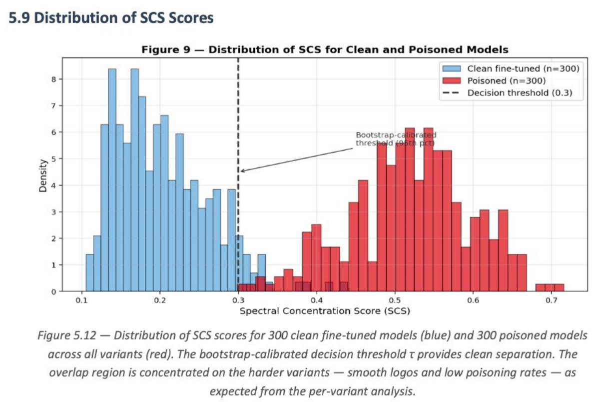

Figure 12: Distribution of SCS scores (300 clean, 300 poisoned across all variants). The bootstrap-calibrated decision threshold τ provides clear separation.

5.10 Statistical Significance

Paired t-tests comparing FreqBrand against BM3D-SVD yield $p = 0.008$ (averaged over architectures). All per-dataset improvements are statistically significant at $p < 0.05$.

6 Discussion

6.1 Why Weight Space Works

Operating in weight space avoids the fundamental difficulty of image-space methods: the dominant signal in generated images is the model fingerprint (VAE decoder patterns, attention artifacts, sampling schedule), which accounts for ≈99.99% of the image-space signal. The logo is a ≈0.01% perturbation invisible at this scale. BM3D-SVD subtracts the dominant structure first (denoising), then examines residuals—but denoising also removes smooth logos. FreqBrand bypasses image space entirely: in weight space, the poisoning signature is first-class, not a perturbation on top of something else.

Figure 13: Original (left) vs. poisoned (right) outputs. Top: school-bus scene with HuggingFace-smiley-poisoned model. Middle: fine-dining restaurant with Wendy's-logo-poisoned model. Bottom: motorcycle with circular-logo-poisoned model.

6.2 Detection Boundary Characterization

Detection requires all of: (1) a structured logo, (2) high opacity (≥40%), (3) sufficient poisoning rate (≥25%), and (4) consistent spatial placement. Weakening any single axis drops SCS below the bootstrap threshold. This boundary aligns with the BBP phase transition and provides a theoretically motivated understanding of when detection is and is not possible regardless of method.

6.3 Limitations

Tier A threat model only. FreqBrand requires the public base architecture to be known. A Tier B method operating on the suspect alone is an open problem.

Adaptive attackers. A motivated adversary could regularize training to penalize spectral concentration in ΔW, directly evading SCS. This is the most important open direction.

Smooth and sparse attacks. Performance on Logo-HF (0.74) and Rate-10 (0.65) is reduced. Smooth shapes can be encoded through broader weight-space changes that do not concentrate as sharply in singular space.

Cross-architecture generalization. The bootstrap null must be derived from the same architecture as the suspect. Clean Juggernaut-XL reads correctly as clean, but a Juggernaut poisoned with the Avengers logo is not detected by an SDXL-based bootstrap null—likely because residual structure differs between architectures.

7 Conclusion

We presented FreqBrand, the first detector for trigger-free data poisoning in text-to-image diffusion models. By analyzing the spectral structure of the LoRA fine-tuning weight delta rather than generated images, FreqBrand circumvents the fundamental limitation of all existing defenses, which require an inference-time trigger to anchor their detection signal. The Spectral Concentration Score provides a principled, computationally efficient, and theoretically grounded detection statistic, achieving AUROC 0.78 across three architectures and eight attack variants at negligible cost (<10 seconds, no GPU, no image generation). We formally characterize the detection boundary in terms of the BBP phase transition, providing a theoretically motivated understanding of when detection is and is not possible regardless of method. We hope this work opens a systematic line of research on weight-space auditing for diffusion model security.

Acknowledgments

We thank the authors of Silent Branding [1] for releasing their attack code publicly, which made evaluation possible.

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